Enrollment No: ____________________

PARUL UNIVERSITY

FACULTY OF ENGINEERING & TECHNOLOGY

B.Tech Mid Semester Exam

Semester: 4th **Date: (31/01/2025)**

Subject Code: 303191251 **Time: (1hr: 30min)**

Subject Name: Probability, Statistics and Numerical Methods **Total Marks: 40**

Sr.No.			Marks	Co/Po	Blooms Taxonomy
Q.1	**(A)**	**Five One line Questions / MCQS.**	**05**		
	(1)	Nature of the binomial random variable X is: (a) Quantitative (b) Qualitative (c) Discrete (d) Continuous		1	Remembering
	(2)	In case of fitting a quadratic curve $y=a+bx+cx^2$, which of the following is not a normal equation? (a) $\Sigma y=na+b\Sigma x+c\Sigma x^2$ (b) $\Sigma x^2y=a\Sigma x^2+b\Sigma x^3+c\Sigma x^4$ (c) $\Sigma xy=n\Sigma a+b\Sigma x^2+c\Sigma x^3$ (d) $\Sigma xy = a\Sigma x+b\ \Sigma x^2+c\Sigma x^3$		1	Understanding
	(3)	Write mean and variance of the Binomial distribution.		2	Applying
	(4)	An unbiased coin is tossed 3 times. the number of elements in the sample space are ____ (a) 6 (b) 8 (c) 10 (d) 9		2	Analyzing
	(5)	For which value of k will F(x) be the probability mass function? X: -1, 0, 1 F(X): 0.6, k, 0.7 (a) 0.3 (b) -0.3 (c) 0 (d) does not exist		4	Evaluating
	(B)	**Five Fill in the blanks, True/False.**	**05**		
	(1)	P(A∪B∪C) = __________		1	Remembering
	(2)	_______ probability distribution is most suitable for the number of accidents take place at a particular spot.		1	Understanding
	(3)	The value of Correlation coefficient is independent of the change of origin but not of scale.______(True/False)		2	Applying
	(4)	The mean of the Poisson distribution is 0.3, then the variance of the mean i.e. V(mean) is equal to _____		2	Analyzing
	(5)	The method which uses latest available values instead of previous iteration values is called ______ (Gauss Jacobi method / Gauss Seidel method)		4	Evaluating
Q.2		Attempt any four (Short Questions)	12		
	(1)	A company manufactures readymade garments. The monthly advertisement cost (in ten lakhs ₹) and the sale of garments (in crore ₹) for the last six months are given below: Monthly Advertisement Cost (in ten lakhs ₹) x: 5, 8, 10, 15, 12, 9 Sakes (in crore ₹) y: 20, 30, 50, 70, 60, 35		1	Remembering

<table>
<tr><td></td><td></td><td>Draw a scatter diagram from the data and discuss about the 1correlation.</td><td></td><td></td><td></td></tr>
<tr><td></td><td>(2)</td><td>32 cards are drawn from a well shuffled pack of 52 cards the2n find probability for the following cases (i) all cards are picture cards (ii) 1 is number card (iii) all cards are not club.</td><td></td><td>1</td><td>Understanding</td></tr>
<tr><td></td><td>(3)</td><td>An unbiased coin is tossed 7 times. Find the probability of getting (i) exactly 5 heads (ii) at least 5 heads. (iii) at most 2 tails.</td><td></td><td>2</td><td>Applying</td></tr>
<tr><td></td><td>(4)</td><td>Find the regression line y on x for the following data:
<table>
<tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr>
<tr><td>y</td><td>160</td><td>180</td><td>140</td><td>180</td><td>200</td></tr>
</table></td><td></td><td>2</td><td>Analyzing</td></tr>
<tr><td></td><td>(5)</td><td>Find a real root of $xe^{x} = 4$, correct upto three decimal places, by using Newton-Raphson’s method.</td><td></td><td>4</td><td>Evaluating</td></tr>
<tr><td>Q.3</td><td></td><td>Attempt any two questions</td><td>08</td><td></td><td></td></tr>
<tr><td></td><td>(1)</td><td>The following data shows number of calls made by 8 sales representative and the and the number of machine sold.
<table>
<tr><td>No. of sales calls</td><td>20</td><td>40</td><td>20</td><td>30</td><td>10</td><td>10</td><td>20</td><td>20</td></tr>
<tr><td>No. of machines sold</td><td>30</td><td>60</td><td>40</td><td>60</td><td>30</td><td>40</td><td>40</td><td>50</td></tr>
</table>
Obtain Karl Pearson’s correlation coefficient and comment on its values.</td><td></td><td>1</td><td>Remembering</td></tr>
<tr><td></td><td>(2)</td><td>Find the constant k such that $f(x) = k(1-x^2)\ ; \quad 0 < x < 4$
$= 0 \quad ;$ otherwise
Is a probability function. Also, find the distribution function F(x) and $P(2<X\leq 3)$, $P(X< 3)$,$P(X\geq 3)$</td><td></td><td>2</td><td>Analyzing</td></tr>
<tr><td></td><td>(3)</td><td>Using Bisection method , find the real root of $xe^{x}-1=0$ correct upto two decimal places.</td><td></td><td>4</td><td>Evaluating</td></tr>
<tr><td>Q.4</td><td>(A)</td><td>Solve the following system of equations by Gauss Jacobi method $6x+2y-z=4$ $x+5y+z=3$ $2x+y+4z=27$ perform only 4 steps.</td><td>05</td><td>2</td><td>Applying</td></tr>
<tr><td></td><td>(B)</td><td>A company is producing a product by three machines A, B and C is 30%, 50% of the product is produced by A, B and the remaining is produced by C. Machine A, B and C produce 3%,5% and 7% defective product. A customer randomly selected a product and it found to be non-defective, find the probability that it is produced by machine C.</td><td>05</td><td>1</td><td>Remembering</td></tr>
<tr><td></td><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td></td><td>(B)</td><td>What is the probability that a standard normal variate Z will be (i) greater than 1.72? (ii) less than -1.33? (iii) lying between -1 and 1? (iv) lying between 1.5 And 2.3? ($P(0<Z<1.72)=0.4564$, $P(0<Z<1.33)=0.4066$, $P(0<Z<1)=0.3413$, $P(0<Z<1.5)=0.4332$, $P(0<Z<2.3)=0.4893$)</td><td>05</td><td>4</td><td>Evaluating</td></tr>
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